FEDERAL SPECIFICATION

COAT, LABORATORY, DISPOSABLE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for disposable laboratory coats.

1.2 Classification. The coats described in this specification shall be made of the following sizes and colors.

1.2.1 Sizes. The coats shall be in the following sizes as specified (see table II and 6.2):

Small
Medium
Large
Extra Large

1.2.2 Color. The coats shall be in the following colors as specified (see 3.3 and 6.2):

White
Blue

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

V-T-276 - Thread, Cotton.
TT-I-542 - Ink, Marking, Laundry, Black.
PPP-B-636 - Boxes, Shipping, Fiberboard.

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Std. No. 751 - Stitches, Seams and Stitchings.
Fed. Std. No. 595 - Colors.
Fed. Test Method Std. No. 191 - Textile Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Association of Textile Chemists and Colorists (AATCC) Standard:

Test Method 22-1967 - Resistance to External (Surface) Wetting: Spray Test.

(Application for copies should be addressed to the American Association of Textile Chemists and Colorists National Headquarters, P.O. Box 886, Durham, NC 27702.)

Technical Association of the Pulp and Paper Industry (TAPPI) Test Methods:

T403 - Bursting Strength of Paper.
T410 - Weight per Unit area (Basis Weight or Substance) of Paper and Paperboard.

(Application for copies should be addressed to the Technical Association of Pulp and Paper Industry, One Dunwoody Park, Atlanta, GA 30341.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material. The material from which the lab coat will be made shall be synthetic or natural fiber or combination thereof and may be scrim or thread reinforced at the manufacturer's option. The material shall conform to the requirements specified in table I.

TABLE I. Material for laboratory coat

Characteristic	Requirements applicable to laboratory coat
Basis weight, pounds/3000 sq. feet, maximum	120
Thickness, inches, maximum	0.020
Bursting strength, pounds per square inch, minimum	16.0
Tensile strength, pounds per inch width, minimum,	
Cross direction	8.0
Machine direction	8.0
Elongation, percent, minimum:	
Cross direction	15
Machine direction	15
Air permeability, $ft^3/min./ft^2$, minimum	10
Abrasion, cycles, weakest direction, dry, minimum	10
Flame resistance:	
After flame, maximum, each principle direction in seconds	3.0
Char length, maximum, each principle direction in inches	9.0
Water resistance, percent, minimum	50

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3.1.1 Toxicity. The material shall contain no known toxic or irritating elements (see 4.1.2.2).

3.1.2 Thread. The cotton machine thread shall be white, type IA3, ticket No. 70, 2 ply or ticket No. 80, 2 ply of V-T-276.

3.1.3 Snaps. The snaps for the front closure of the lab coat shall be white. There shall be at least 3 snaps.

3.1.4 Marking. Each garment shall be labeled according to proper size. The size shall be represented by "S" for small, "M" for medium, "L" for large, and "XL" for extra large. The letters shall be no smaller than 1/2 inch and no larger than 1-1/2 inches. The method of labeling shall be stamped, stenciled or a pressure sensitive label. This marking shall be centered on the inside back of the coat not more than 2 inches below the collar and neckline seam.

3.2 Design. The laboratory coat shall have a snap front closure, lapel, long raglan sleeves, a left chest pocket and larger lower pocket on the right. The sleeves and coat bottom shall be unhemmed.

3.2.1 Sizes. The coats shall be in all of the sizes in table II.

TABLE II. Table of sizes

Laboratory Coat	Small	Medium	Large	Extra Large
Ladies Size	6-8	10-12	14-16	18-20
Mens Size	34-36	38-40	42-44	46-48

3.3 Colors. The coats shall be one of the following colors and be approximately similar to the color indicated in Fed. Std. No. 595.

White - 27722
Blue - 35488 or 35550

3.4 Construction.

3.4.1 Stitches, seams, stitching. The stitching for the laboratory coat shall conform to type 301, type 304, type 504, or type 512 of Fed. Std. No. 751. The stitches shall be not less than 6 per inch. All joining seams shall be formed 1/2 inch from the edge of the material.

3.4.2 Thread breaks and ends of seams. Ends of all seams and stitching produced with 301 stitch type, when not caught in other seams or stitching, shall be backtacked not less than 1/4 inch. Ends of a continuous line of stitching shall overlap not less than 1/2 inch. Thread breaks (all stitch types) shall be secured by stitching back of the break by at least 1/2 inch. Skipped stitches or thread breaks on 304 stitch type may be repaired using 301 stitch type.

3.5 Workmanship. The finished laboratory coat shall conform to the quality of the product established in this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Sampling for examination and tests shall be performed in accordance with MIL-STD-105, except where otherwise indicated. For purposes of sampling, a lot shall consist of all laboratory coats of the same size and color submitted for inspection at one time.

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4.1.2 Inspection of components and materials. In accordance with 4.1, the supplier is responsible for insuring that components and materials were manufactured, examined, and tested in accordance with this specification or referenced subsidiary specifications and standards.

4.1.2.1 Component testing. Each lot of material used in the fabrication of the laboratory coats shall be tested in accordance with the applicable test methods specified hereinafter to determine compliance with the requirements of section 3. Each inspection lot shall consist of a component of a single type or grade manufactured by the same source, under essentially the same conditions and used in the production of the end item submitted for inspection at one time.

4.1.2.2 Evidence of compliance. In lieu of his own facilities, the supplier may submit a report of analysis of component materials from a laboratory acceptable to the Government whether an independent test facility or the laboratory of the raw material supplier, as evidence of compliance with the specification requirements. The report of analysis shall include, in addition to the actual data, the manufacturer's name, name and address of the testing laboratory, date of test, and quantity of material represented by the test report. The toxicity requirement shall be satisfied by a supplier's certificate of compliance.

4.2 Testing of components. The laboratory coat material shall be tested for the applicable characteristics specified in table III. Tests shall be made in accordance with the indicated method as referenced in the table. The lot size shall be expressed in pounds of material. The sample unit shall consist of not less than 3000 square inches of material. The sample size shall be drawn in accordance with level S-1 of MIL-STD-105. Test results shall include all values upon which the results are based and there shall be no failures to meet the sample unit or lot average requirements. All samples shall be conditioned and tested under uniform atmospheric conditions as specified in TAPPI method T402.

TABLE III. Instructions for testing components

Characteristic	Specification reference: Requirement	Specification reference: Test Method	Requirements applicable to: Individ. unit	Requirements applicable to: Lot aver.	Number determinations per sample unit	Results reported numerically to nearest
Basis weight	3.1	410[1]		X	1 on 100 sq. in specimen	0.1 pound
Bursting strength	3.1	403[1]		X	10 (5 on each side)	1 point
Thickness	3.1	5030[1,2,3]		X	Average of 5	.0001 inch
Tensile strength:						
Cross direction	3.1	4.4.1		X	Average of 5	0.5 pound
Machine direction	3.1	4.4.1		X	Average of 5	0.5 pound
Elongation:						
Cross direction	3.1	4.4.1		X	Average of 5	1 percent
Machine direction	3.1	4.4.1		X	Average of 5	1 percent
Air permeability	3.1	5450.1[2]		X	Average of 5	0.1 cu. ft./ min./ft.[2]
Flame resistance:						
After flame	3.1	5903[2]	X		Average of 5	1/5 second
Char length	3.1	5903.2[2,4]	X		Average of 5	0.1 inch
Abrasion resistance:						
Dry	3.1	4.4.2		X	Average of 10	1 cycle
Water resistance	3.1	22-1967[5]	X		6 (3 one each side)	1 second

1 Indicates test method of TAPPI Test Method.
2 Indicates test method of Federal Test Method Standard No. 191, except that the number of determinations per sample unit shall be as specified in this table.
3 Procedure for nonwoven material shall be as specified in this table.
4 The 0.25 pound weight shall be used.
5 Indicates test method of American Association of Textile Chemists and Colorists.

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4.3 Inspection of the end item.

4.3.1 Examination of preparation for delivery. An examination shall be made to determine that packing and marking comply with the requirements in Section 5. Defects shall be as specified below. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL 4.0, expressed in terms of defects per hundred units.

Examine	Defects
Packing	Not level specified. Not in accordance with contract requirements. Container material not as specified. Closures not accomplished by specified or required methods.
Marking	Marking illegible, incorrect, incomplete, omitted, or not in accordance with contract requirements.

4.3.2 Examination for visual defects. The sample unit for this examination shall be one lab coat. The inspection level shall be S-2 with an AQL of 2.5, expressed in terms of percent defective.

Examine	Defects
Appearance	Not clean; any discoloration or mark. Extreme variation in shade of the component parts. Any hole, break, burn, cut, tear, or needle chew.
Components	Any component part missing.
Seams and stitching	Seam or stitch type not as specified. Any part of garment caught in any unrelated operation or stitching. Thread break(s) or end(s) not backstitched or not caught in other seams when required. Ends of a continous line of stitching not overlapped or overlapped less than 1/2 inch. Stitches skipped or broken for distance of 1/2 inch or more. Looper thread of 401 stitch type exposed on outside of garment. Any raw edge where edge is required to be turned under. Seam loose, puckered, or pleated. Stitching not specified distance from edge of material.
Stitches	Number of stitches per inch more or less than specified.
Snaps	Any snap missing, loose, broken, or otherwise defective.
Marking	Omitted; illegible, incorrect, or not properly positioned.
Count	The lot shall be unacceptable if the average count per shipping container for all containers examined is less than specified or indicated.

4.4 Tests.

4.4.1 Tensile strength and elongation. Tensile strength and elongation are determined in accordance with method 5100 of Fed. Test Method Standard 191. The tensile strength and elongation of a specimen are determined simultaneously.

4.4.2 Abrasion resistance. The abrasion resistance shall be performed in accordance with method 5306 of Fed. Test Method Standard No. 191, using a CS10 Calibrase wheel and a 500 gram load. The end point shall be that point at which the first hole is worn completely through the specimen.

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5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. Coats, of one description, shall be folded to afford a minimum of creases, compressed and packed in a box conforming to PPP-B-636, class weather-resistant. Closure shall be in accordance with the appendix to the box specification.

5.1.2 Level B. Twenty-five (25) coats, of one description, shall be individually folded to afford a minimum of creases, compressed and packed in a box conforming to PPP-B-636, class domestic. Closure shall be in accordance with the appendix to the box specification.

5.1.3 Level C. Coats of one description shall be packed in containers to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules, as applicable. The supplier may use his standard practice when it meets this requirement.

5.2 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123 or MIL-STD-129, as applicable.

6. NOTES

6.1 Intended Use. This disposable laboratory coat is intended for use as a protective covering for street clothing when conditions warrant.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Size and color of coat (see 1.2).
(c) Selection of applicable levels of packing (see 5.1).
(d) Quantity per shipping container.

Preparing Activity: GSA - FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 30 cents each.

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